In Situ Gamma-Ray Spectrometric Mapping of Beach Sand Radioactivity in Niterói, Brazil

ID:G02_060

XXV ISSSD
XXV International Symposium on Solid State Dosimetry
September 23-26, 2025 | Belo Horizonte – Brazil

G. A. S. Rodé[1,3]*, I. B. F. Martins[1], T. A. Lima[2], R. C. Florenzano[1], R. C. Nunes[3], P. A. L. Cruz[3], E. M. Oliveira[3], E. M. Souza[4], J. L. Talhofer[1,2]

[1]Federal Fluminense University ; [2] Federal University of Rio de Janeiro; [3] Institute of Nuclear Engineering; [4] State University of Rio de Janeiro

***Correspondence to: gabrielarodee@gmail.com**

Introduction

External human exposure to radiation mainly originates from natural sources, particularly radionuclides from the ^{238}U, ^{232}Th, and ^{40}K series present in soils and rocks. Brazil holds some of the world's largest deposits of these minerals, and coastal sands often contain variable concentrations of uranium- and thorium-associated minerals. Such mineralogical associations may influence exposure of beach users.

Icaraí Beach (Niterói, RJ), although unsuitable for swimming, has one of the most intensively used sand areas in the region. To support public health protection, this study aimed to perform an in situ radiometric mapping of sand along Icaraí Beach to assess exposure levels and radionuclide distribution.

Methodology

Radiation levels were measured *in situ* using a portable gamma-ray spectrometer (Atomtex AT1125A), equipped with a NaI(Tl) scintillation detector and a Geiger–Müller tube, suitable for X- and gamma-radiation monitoring. The device has a detection range from 30 nSv/h to 100 µSv/h, covering an energy window of 50 keV to 3 MeV, with internal data storage and PC transfer capacity, allowing precise recording of radiometric data under field conditions.

Before fieldwork, a digital georeferencing survey of the study area was carried out using Google Earth, which enabled the definition of sampling points and optimization of the logistics. Monitoring was strategically centered on two stormwater outfalls (points 01 and 11), considered potential hotspots of anthropogenic influence. Sampling points were then distributed systematically along the beach, with 50 m spacing in areas adjacent to the drains and 200 m spacing across the central sector, ensuring adequate spatial coverage of radiometric variability.

At each location, measurements were subdivided into up to three distinct transects to capture environmental heterogeneity: (1) Shoreline, (2) Sand Strip 1, and (3) Sand Strip 2, positioned at intervals of approximately 10–30 m from each other. This approach was designed to reflect the dynamic nature of sediment deposition and radionuclide distribution along the beach profile, thereby providing more representative data on both natural background radiation and potential localized anomalies.

Figure 1 - Fieldwork images, by Iris Martins.

Results

Higher dose rates were recorded at points 01 and 11, near stormwater and sewage outfalls, and at points 02 and 13, likely influenced by waste dispersion and heavy mineral accumulation. In contrast, points 05 and 06 showed the lowest values, suggesting reduced external exposure in those areas (Table 1).

All results remain below the ICRP (1990) public dose limit of 1 mSv/year, indicating no significant radiological risk. However, measured doses (0.204–0.289 mSv/year) exceed the global average of 0.07 mSv/year (UNSCEAR, 2000), reflecting the influence of local geology on radiation levels.

Table 1 - Ambient Equivalent Dose Rate and Estimated Mean Annual Effective Dose at Sampling Locations.

Sampling Point	Shoreline (nSv/h)	Sand Strip 1 (nSv/h)	Sand Strip 2 (nSv/h)	Estimated Mean Annual Effective Dose (mSv/year)
Point 01	33 ±12%	-	-	0.28908
Point 02	34 ±16%	30 ±12%	-	0.28032
Point 03	24 ±12%	29 ±16%	29 ±20%	0.23944
Point 04	23 ±20%	27 ±16%	26 ±14%	0.22192
Point 05	21 ±12%	23 ±12%	26 ±20%	0.20440
Point 06	21 ±16%	21 ±16%	28 ±16%	0.20440
Point 07	20 ±16%	20 ±13%	33 ±12%	0.21316
Point 08	22 ±20%	21 ±20%	30 ±15%	0.21316
Point 09	27 ±20%	27 ±20%	28 ±10%	0.23944
Point 10	23 ±20%	24 ±15%	28 ±20%	0.21900
Point 11	26 ±20%	31 ±20%	35 ±20%	0.26864
Point 12	24 ±20%	24 ±12%	-	0.21024
Point 13	33 ±20%	-	-	0.28908

Conclusion

Dose rates along Icaraí Beach ranged from 20–35 nSv/h, corresponding to 0.204–0.289 mSv/year. Although below the limit of 1 mSv/year, these values exceed the global average of 0.07 mSv/year. Higher doses occurred near stormwater outfalls, likely due to radionuclide accumulation, while lower values were observed in less impacted areas. Results indicate no significant radiological risk but emphasize the importance of continuous monitoring in urban beaches with intense public use.

Acknowledgements

CAPES
FAPERJ Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro
CNPq Conselho Nacional de Desenvolvimento Científico e Tecnológico

References